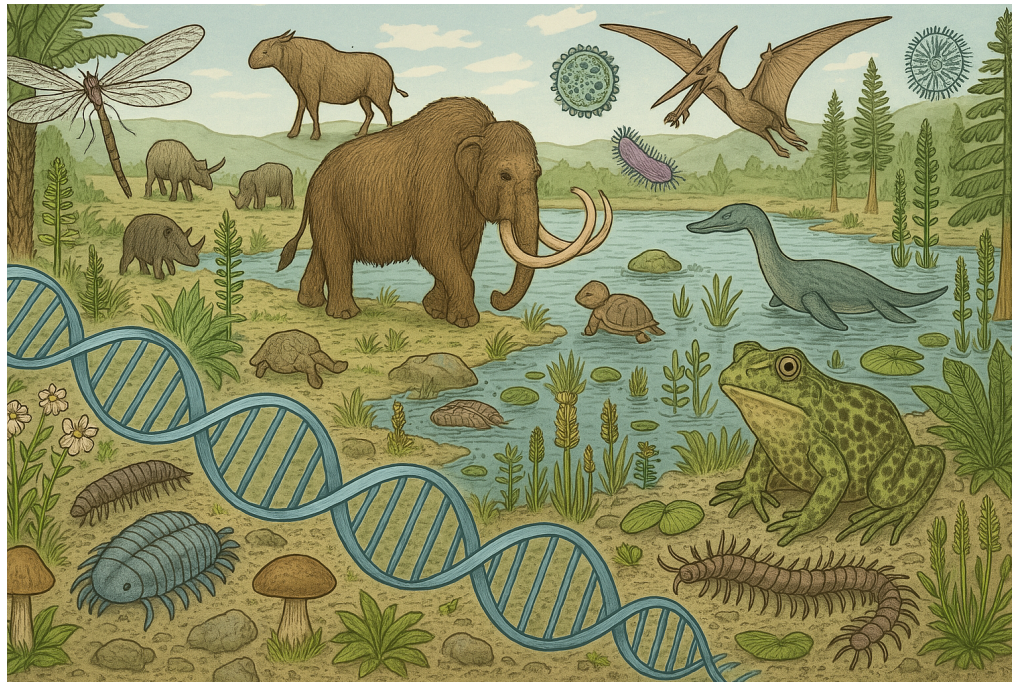


ANCIENT METAGENOMICS

Physalia course, online, 27-29 May 2025

Metagenome *de-novo* assembly and quality control

Nikolay Oskolkov, Lund University, NBIS SciLifeLab



Typical analysis methods used in metagenomics

1) Alignment:



2) Classification:



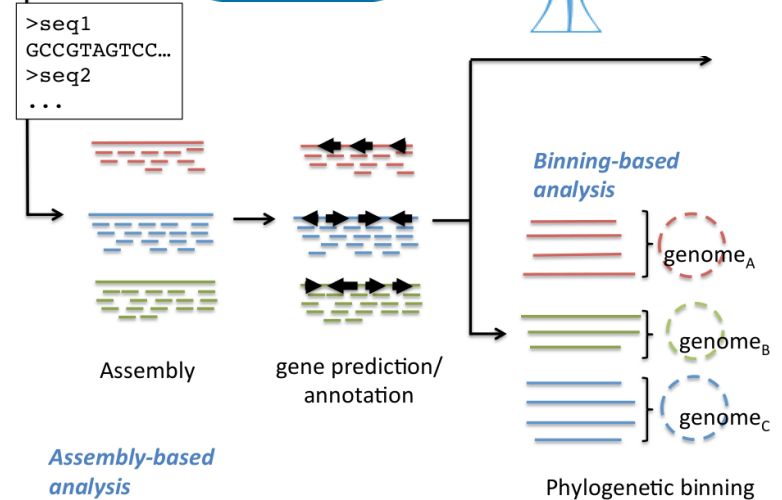
Centrifuge

MetaPhlan

Clark

Reference based:
assume similarity to reference

3) De-novo assembly:

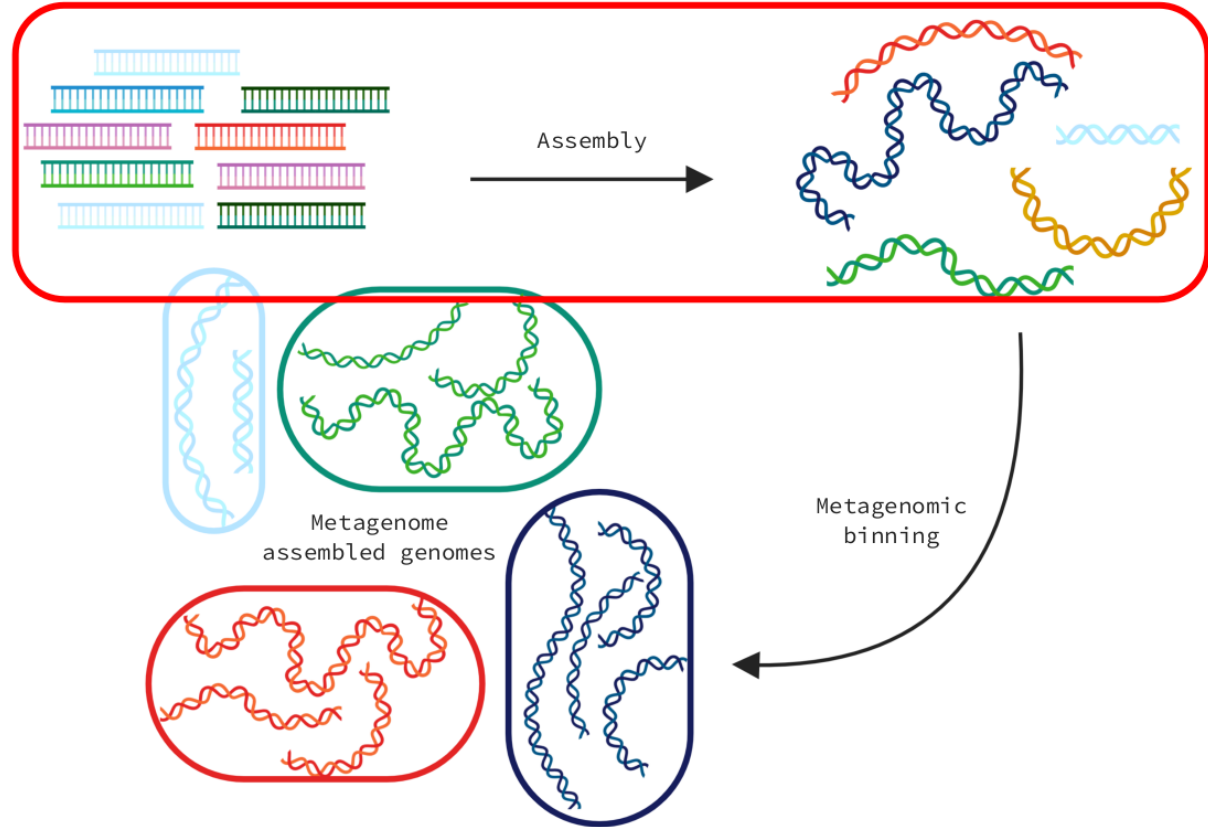


Reference free:
unbiased but challenging

De novo assembly

Assemble short nucleotide sequences into longer sequences by finding their overlap / consensus without reference genome

- No reference available
- Uneven and complex communities



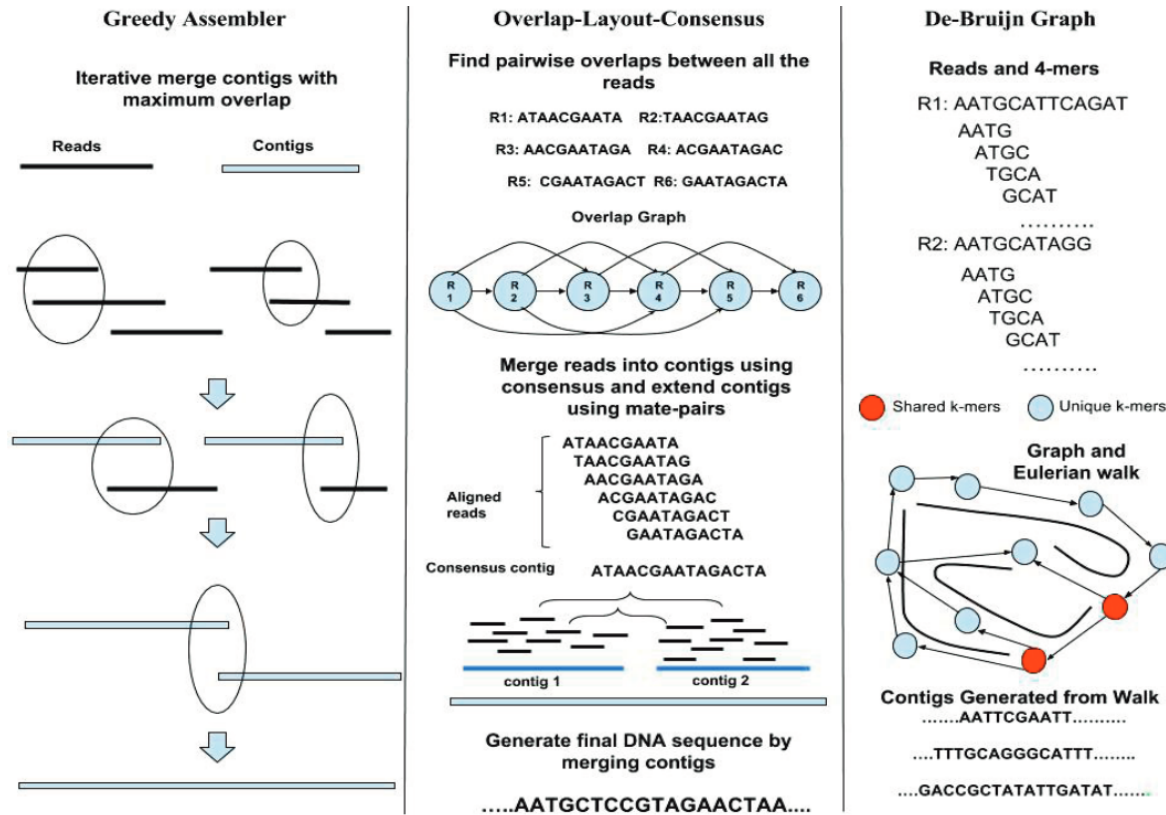


Figure 1: Overview of different de novo assembly paradigms. Schematic representation of the three main paradigms for genome assembly – Greedy, Overlap-Layout-Consensus, and de Bruijn. In Greedy assembler, reads with maximum overlaps are iteratively merged into contigs. In Overlap-Layout-Consensus approach, a graph is constructed by finding overlaps between all pairs of reads. This graph is further simplified and contigs are constructed by finding branch-less paths in the graph, and taking the consensus sequence of the overlapping reads implied by the corresponding paths. Contigs are further organized and extended using mate pair information. In de Bruijn graph assemblers, reads are chopped into short overlapping segments (k-mers) which are organized in a de Bruijn graph structure based on their co-occurrence across reads. The graph is simplified to remove artifacts due to sequencing errors, and branch-less paths are reported as contigs.

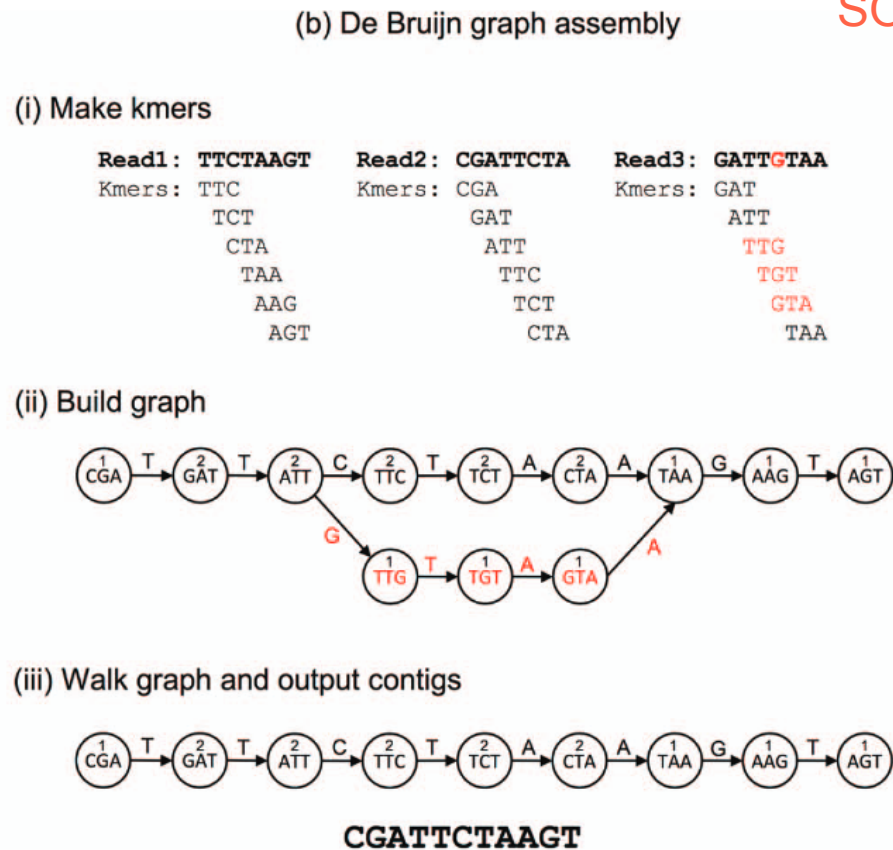
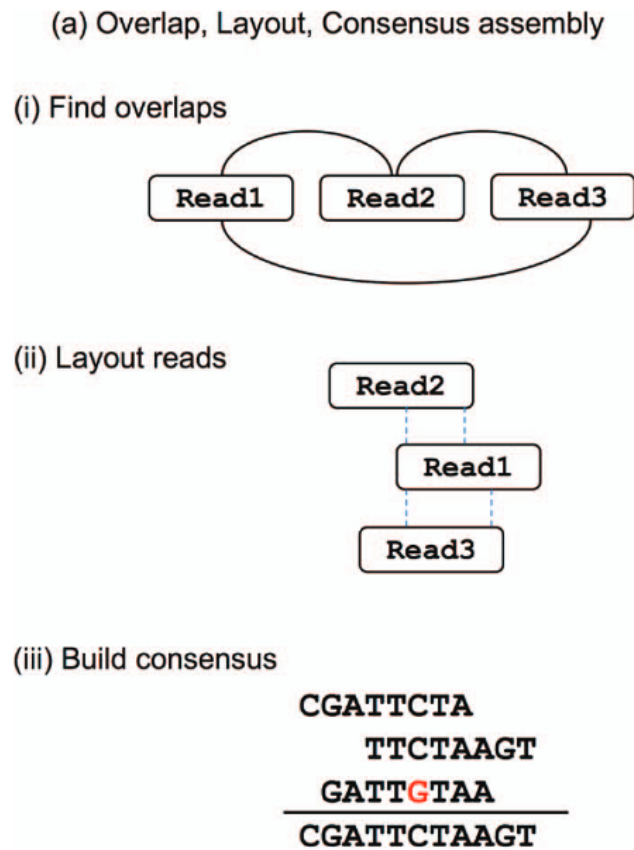


Figure 1. Two different approaches to genome assembly: (a) in Overlap, Layout, Consensus assembly, (i) overlaps are found between reads and an overlap graph constructed (edges indicate overlapping reads). (ii) Reads are laid out into contigs based on the overlaps (dashed lines indicate overlapping portions). (iii) The most likely sequence is chosen to construct consensus sequence. (b) In dBg assembly, (i) reads are decomposed into kmers by sliding a window of size k across the reads. (ii) The kmers become vertices in the dBg, with edges connecting overlapping kmers. Polymorphisms (red) form branches in the graph. A count is kept of how many times a kmer is seen, shown here as numbers above kmers. (iii) Contigs are built by walking the graph from edge nodes. A variety of heuristics handle branches in the graphs—for example, low coverage paths, as shown here, may be ignored.

Popular de Bruijn graph *de-novo* metagenomic assemblers for short Illumina reads

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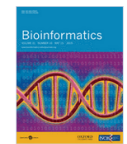
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Volume 31, Issue 10
May 2015

Article Contents

Abstract

1 Introduction

2 Methods

3 Results

4 Conclusions

Acknowledgements

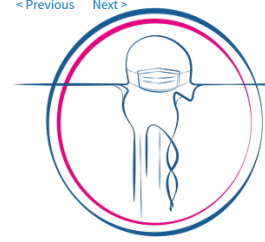
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Supplementary data

< Previous Next >



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JOURNAL ARTICLE

MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct *de Bruijn* graph

Dinghua Li, Chi-Man Liu, Ruibang Luo, Kunihiro Sadakane, Tak-Wah Lam
Author Notes

Bioinformatics, Volume 31, Issue 10, May 2015, Pages 1674–1676, <https://doi.org/10.1093/bioinformatics/btv033>

Published: 20 January 2015 Article history ▾

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Abstract

Summary: MEGAHIT is a NGS *de novo* assembler for assembling large and complex metagenomics data in a time- and cost-efficient manner. It finished assembling a soil metagenomics dataset with 252 Gbps in 44.1 and 99.6 h on a single computing node with and without a graphics processing unit, respectively. MEGAHIT assembles the data as a whole, i.e. no pre-processing like partitioning and normalization was needed. When compared with previous methods on assembling the soil data, MEGAHIT generated a three-time larger assembly, with longer contig N50 and average contig length; furthermore, 55.8% of the reads were aligned to the assembly, giving a fourfold improvement.

Availability and implementation: The source code of MEGAHIT is freely available at <https://github.com/voutcn/megahit> under GPLv3 license.



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metaSPAdes: a new versatile metagenomic assembler

Sergey Nurk^{1,4}, Dmitry Meleshko^{1,4}, Anton Korobeynikov^{1,2} and
Pavel A. Pevzner^{1,3}

Author Affiliations

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⁴ These authors contributed equally to this work.

Abstract

While metagenomics has emerged as a technology of choice for analyzing bacterial populations, the assembly of metagenomic data remains challenging, thus stifling biological discoveries. Moreover, recent studies revealed that complex bacterial populations may be composed from dozens of related strains, thus further amplifying the challenge of metagenomic assembly. metaSPAdes addresses various challenges of metagenomic assembly by capitalizing on computational ideas that proved to be useful in assemblies of single cells and highly polymorphic diploid genomes. We benchmark metaSPAdes against other state-of-the-art metagenome assemblers and demonstrate that it results in high-quality assemblies across diverse data sets.

Metagenome sequencing has emerged as a technology of choice for analyzing bacterial populations and the discovery of novel organisms and genes (Tyson et al. 2004; Venter et al. 2004; Yooseph et al. 2007; Arumugam et al. 2011). In one of the early metagenomics studies, Venter et al. (2004) attempted to assemble the complex Sargasso Sea microbial community but, as the study stated, failed. On the other side of the spectrum of metagenomics studies, Tyson et al. (2004) succeeded in assembling a simple microbial community consisting of a few species.

These landmark studies (Tyson et al. 2004; Venter et al. 2004) used conventional assembly tools—namely, Celera (Myers et al. 2000) and JAZZ (Aparicio et al. 2002)—with minor modifications. Since they were published, many specialized metagenomic assemblers have been developed (Koren et al. 2011; Laserson et al. 2011; Peng et al. 2011, 2012; Boisvert et al. 2012; Namiki et al. 2012; Haider et al. 2014; Li et al. 2016). However, bioinformaticians are still struggling to bridge the gap between assembling simple and complex microbial communities (for a review see Gevers et al. 2012). Meanwhile, many

« Previous | Next Article »
Table of Contents

This Article

Published in Advance March 15, 2017, doi:
10.1101/gr.213959.116
Genome Res. 2017, 27: 824–834
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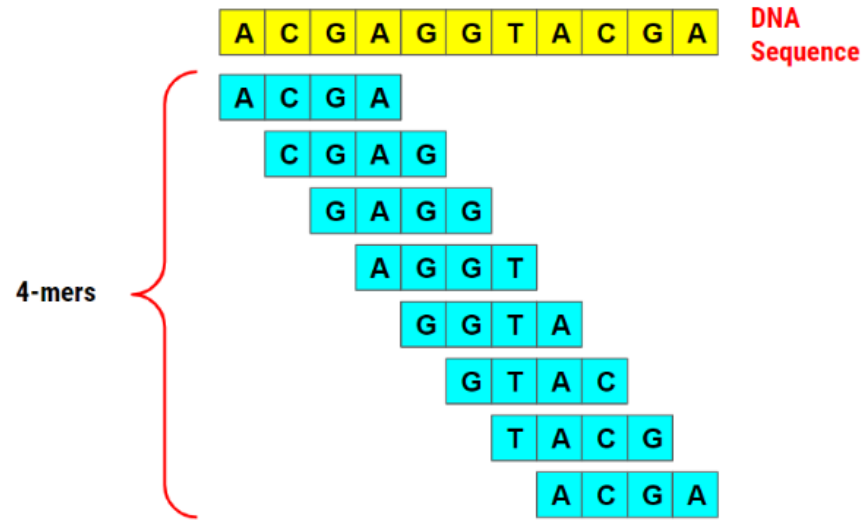
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We are going to use Megahit in the exercises

De novo assembly using MEGAHIT

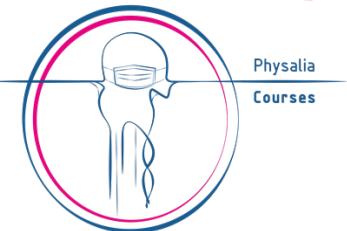
MEGAHIT: de Bruijn-graph assembler using a distribution of different k-mer lengths inferred from the length of the sequencing data



reasons for using MEGAHIT:

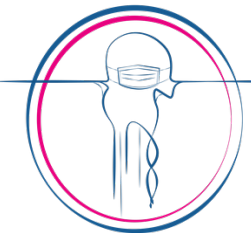
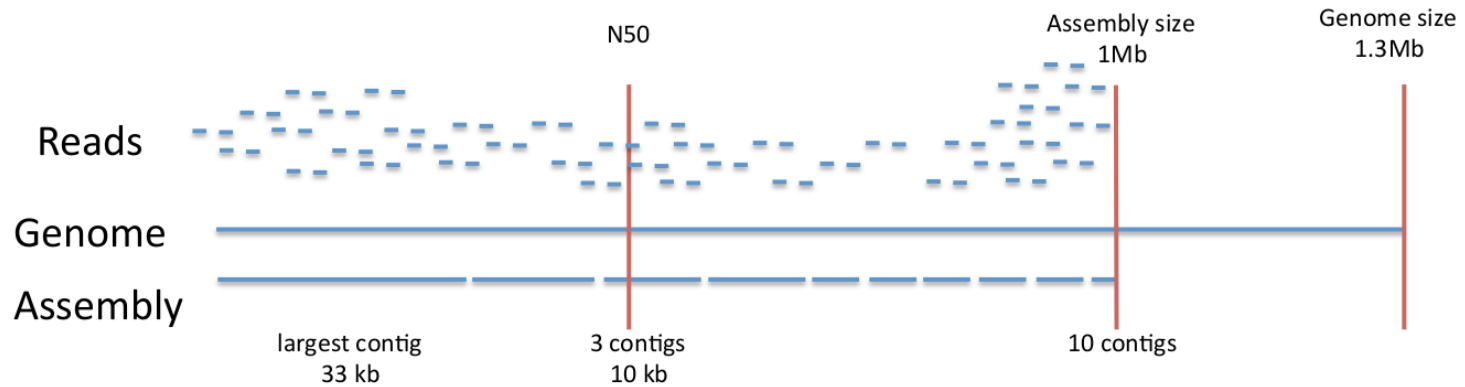
- **low-memory** footprint
- has little issues with the **presence of ancient DNA damage**
- works with **single-end data**

BUT: lower assembly quality than other assemblers for modern sequencing data (see CAMI II challenge; DOI: [10.1038/s41592-022-01431-4](https://doi.org/10.1038/s41592-022-01431-4))



Assembly metrics

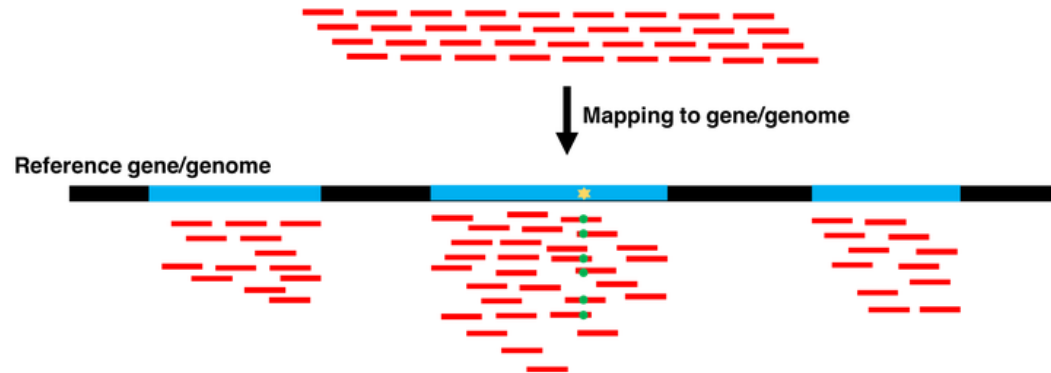
- assembly size
- number of contigs, largest contig
- N50



Alignment against the contigs

Many of the following steps require the alignment of the short-read data against the de novo assembled contigs, e.g.

- correction of the contig sequences
- binning of the contigs into MAGs (coverage along the contigs)
- quantification of the presence of ancient DNA damage



Taxonomic classification - on contig level

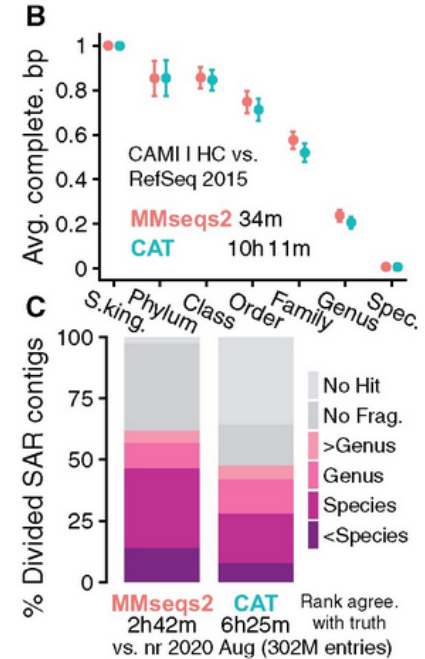
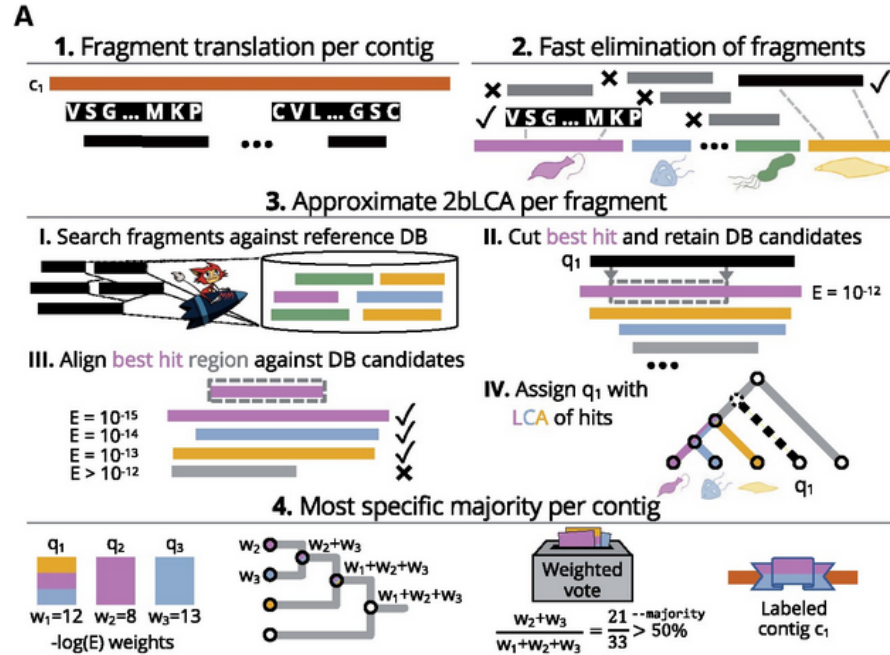
The likely taxonomic origin of contigs can be determined by aligning them against a reference database.

available aligners:

- BLAST/DIAMOND
- Kraken2
- Centrifuge
- MMSeqs2

available databases:

- NCBI NT/RefSeq
- GTDB



Mirdita *et al.* (Bioinformatics, 2021; doi: bioinformatics/btab184) Fig. 1

Authentication of *de-novo* assembled contigs

pyDamage evaluates the **amount of aDNA damage** and **tests the hypothesis** whether the model assuming presence of aDNA damage better explains the data than a null model



PyDamage: automated ancient damage identification and estimation for contigs in ancient DNA *de novo* assembly

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²Faculty of Biological Sciences, Friedrich-Schiller Universität Jena, Jena, Germany

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⁵Department of Anthropology, Harvard University, Cambridge, MA, United States of America

ABSTRACT

DNA *de novo* assembly can be used to reconstruct longer stretches of DNA (contigs), including genes and even genomes, from short DNA sequencing reads. Applying this technique to metagenomic data derived from archaeological remains, such as paleofeces and dental calculus, we can investigate past microbiome functional diversity that may be absent or underrepresented in the modern microbiome gene catalogue. However, compared to modern samples, ancient samples are often burdened with environmental contamination, resulting in metagenomic datasets that represent mixtures of ancient and modern DNA. The ability to rapidly and reliably establish the authenticity and integrity of ancient samples is essential for ancient DNA studies, and the ability to distinguish between ancient and modern sequences is particularly important for ancient microbiome studies. Characteristic patterns of ancient DNA damage, namely DNA fragmentation and cytosine deamination (observed as C-to-T transitions) are typically used to authenticate ancient samples and sequences, but existing tools for inspecting and filtering aDNA damage either compute it at the read level, which leads to high data loss and lower quality when used in combination with *de novo* assembly, or require manual inspection, which is impractical for ancient assemblies that typically contain tens to hundreds of thousands of contigs. To address these challenges, we designed PyDamage, a robust, automated approach for aDNA damage estimation and authentication of *de novo* assembled aDNA. PyDamage uses a likelihood ratio based approach to discriminate between truly ancient contigs and contigs originating from modern contamination. We test PyDamage on both on simulated aDNA data and archaeological paleofeces, and we demonstrate its ability to reliably and automatically identify contigs bearing DNA damage characteristic of aDNA. Coupled with aDNA *de novo* assembly, PyDamage opens up new doors to explore functional diversity in ancient metagenomic datasets.

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Additional Information and

Declarations can be found on

page 16

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